MILITARY SPECIFICATION

FILTER UNIT, NOTCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the technical requirements for High Frequency (HF) dual-channel Notch Filter Unit. This unit shall provide special purpose shipboard receivers with protection from overload by local onboard transmitters over the frequency range of 2.0 to 24.0 Megahertz (MHz).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	-	Shock Test, H. I. (High-Impact); Shipboard Machinery, Equipment and Systems Requirement for
MIL-E-16400	-	Military Specification, Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore; General Specification for
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-C-24056	-	Cabinets, Electronic Equipment (Ships)

STANDARDS

MILITARY

MIL-STD-108	-	Definition of and Basic Requirements for Enclosure for Electric and Electronic Equipment
MIL-STD-167/1	-	Mechanical Vibration of Shipboard Equipment (Type I-Environmental and Type II-Internally Excited)
MIL-STD-242	-	Electronic Equipment Parts
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for Equipment

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MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-749	-	Non-Standard Parts
MIL-STD-756	-	Reliability Prediction
MIL-STD-781	-	Reliability Tests
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1364	-	Standard General Purpose Electronic Test Equipment

HANDBOOKS

MIL-HDBK-472	-	Maintainability Predictions

(Copies of specification, standards, drawings, and publications required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SANDERS ASSOCIATES INC.

DRAWINGS

1047702	-	Notch Filter Assembly, Installation of

(Application for copies should be addressed to Sanders Associates, Incorporated, 95 Canal St., Nashua, N. H. 03060.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (See 6.3).

3.1.1 General characteristcs. The equipment specified herein shall conform to MIL-E-16400 unless otherwise specified.

3.1.2 Description. The notch filter unit consists of a set of eight assemblies connected in series to form a dual-channel rejection network. Each assembly consists of two separate and identical notch filters mechanically tunable over certain frequency ranges as specified herein. Each assembly shall be provided with a switch that allows the dual filters to be bypassed or inserted in series between a shipboard antenna and a receiver input.

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3.1.3 Item diagram. Figure 1 shows the block diagram for two notch filter assemblies. The assembly on the left is an example of bypassed filters. The other assembly is inserted into the receiver input path via the filter on-off switch.

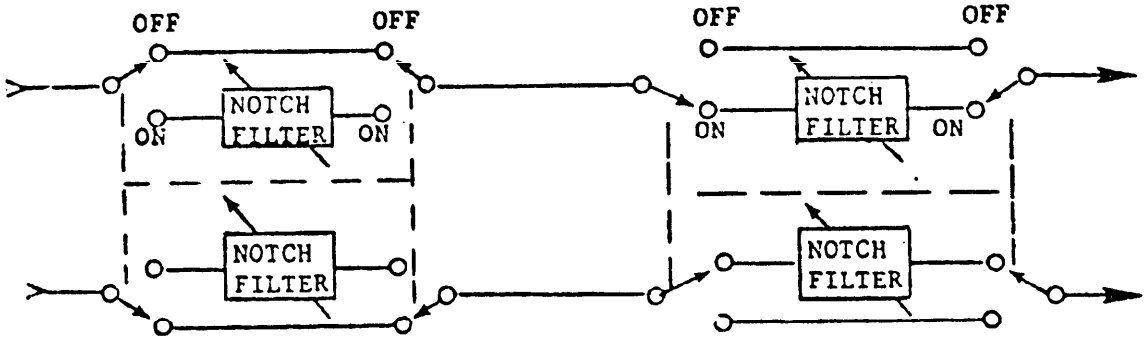


Figure 1. Basic diagram of two notch filter assemblies in a filter unit.

3.2 Electrical characteristics.

3.2.1 Frequency range. Each filter shall have a notch rejection network that is mechanically tunable over one of the following frequency bands:

Band	Frequency (MHz)
1	2.0 - 4.0
2	4.0 - 8.0
3	8.0 - 16.0
4	16.0 - 24.0

3.2.2 Filter complement. The eight dual-channel filter assemblies in each notch filter unit shall be distributed over the frequency bands in the following manner:

Band	No. of assemblies
1	2
2	2
3	3
4	1

3.2.3 Notch attenuation. Attenuation of the notch filter assemblies, excluding insertion loss, shall have the following response characteristics:

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Band	Bandwidth about notch center frequency for attenuation:			
	60 dB	20 dB	3 dB	1 dB
1	±5 kHz min.	±100 kHz min.	±350 kHz max.	±500 kHz max.
2	±5 kHz min.	±100 kHz min.	±350 kHz max.	±500 kHz max.
3	±5 kHz min.	±100 kHz min.	±1300 kHz max.	±1600 kHz max.
4	±5 kHz min.	±100 kHz min.	±1900 kHz max.	±2400 kHz max.

3.2.4 Insertion loss. With eight assemblies connected in series and in the bypass mode, the attenuation of an radio frequency (RF) signal between 0.1 and 32.0 MHz shall not be more than 1.0 decibel (dB) in each channel. Each filter in the series shall not contribute more than 0.2 dB of this figure. With eight assemblies connected in series and switched in, the attenuation of an RF signal between 0.1 and 32.0 MHz shall not be more than 3.0 dB in each channel, except within the notch response characteristics of 3.2.3. Filters in the series shall not contribute more than 0.7 dB of this figure.

3.2.5 Phase tracking. Reference phase delays shall be established for each band of notch filter assemblies and associated cables and for the bypass mode (all notch filter assemblies bypassed) of the notch filter unit. The first article and follow-on production units shall be calibrated (by adjusting circuit delay taps, cable lengths, and so forth) to maintain a phase tracking tolerance of ±0.1 electrical degree with respect to the established reference phase delays at a reference frequency of 15.0000 MHz between similar notch filter bands and notch filter units in the bypass mode. Measurements shall be made within a temperature range of 25° ±5° centigrade (C). Reference cable lengths shall be established for cables associated with switching functions within the notch filter units. Cable lengths shall be cut within ±0.25 inches (6.35mm) of the established reference lengths for corresponding cable switching functions. Verification of compliance with the established reference phase delays and reference cable lengths shall be included in the production acceptance test procedures. Typical filter and cable delay considerations are depicted in figure 2.

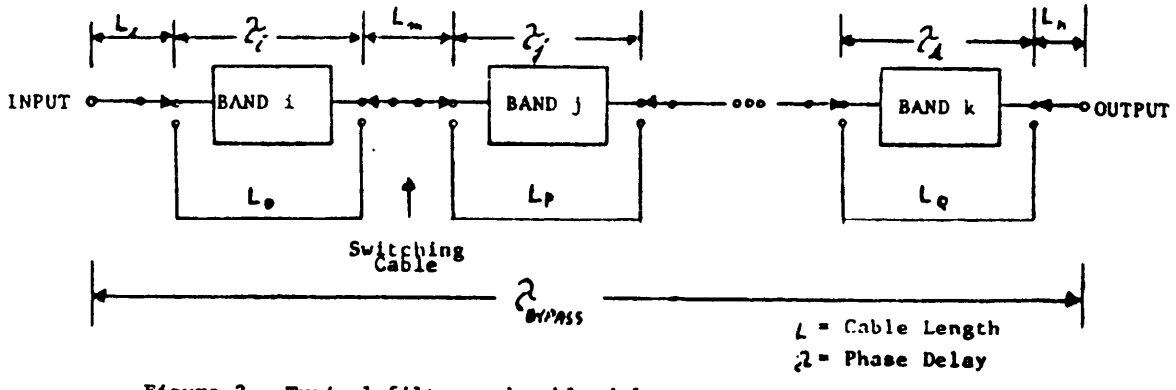


Figure 2. Typical filter and cable delays.

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3.2.6 Voltage standing wave ratio (VSWR). With the notch filter unit output terminated into a 50 ohm resistive load, the voltage standing wave ratio shall not be more than 2.0 to 1.0 out of the notch response between 0.1 and 32.0 MHz. The VSWR within the notch one dB points shall not be more than 3.0 to 1.0. Any combination up to eight assemblies may be switched in for the above tests.

3.2.7 RF input power. Each filter shall conform to specification with up to 20 watts rms of RF power applied to the input between 0.1 and 32.0 MHz.

3.2.8 Channel isolation. The isolation between the filters of an assembly shall not be less than 80 dB between 0.1 and 32.0 MHz.

3.2.9 Intermodulation susceptibility. Spurious responses and intermodulation products in each filter shall be no greater than -50 decibel referenced to one milliwatt (dBm) at the filter output for two incident signals, one watt each, located outside the notch response and between 0.1 and 32.0 MHz.

3.2.10 Tuning. Each filter shall be tuned by concentric, integral coarse and fine tuning controls. The coarse tuning knob shall be able to continuously tune the filter through a frequency band in less than one revolution. The fine tuning knob shall be able to tune the filter through a band with not less than five revolutions. A minimum of seventeen frequency markings per band shall be provided that indicate the tuned frequency within an accuracy of ±5 percent. The dial shall be at least 2.5 inches (6.35cm) across. Tuning controls for both filters of an assembly may be ganged; however, the center frequencies of the notches shall track within ±2 kilohertz (kHz).

3.2.11 Channel notch alignment. A momentary make-before-break switch shall be provided by the notch filter unit that reverses the channel A output of the assembly with the channel B output to permit nulling RF signals using one receiver (see figure 3).

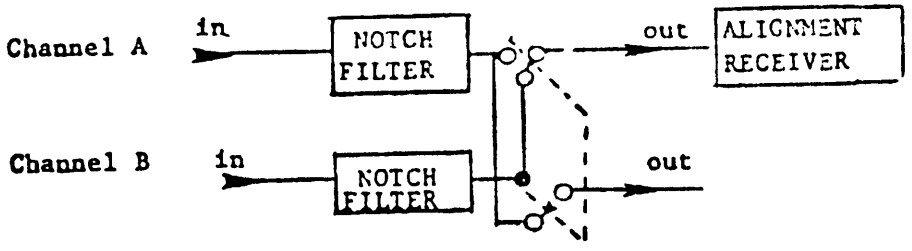


Figure 3. Channel notch alignment.

3.2.12 Filter in/out. A switch shall be provided for switching the assembly in or out (see figure 1). A white indicator light shall be activated when the assembly is switched in.

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3.2.13 DC electrical power. Indicator lights of the notch filter unit shall use ±15 volts direct current (VDC) with a maximum current requirement of 1.5 amps. Loss of electrical power shall only affect the operation of the indicator lights and not the filter operation.

3.2.14 Cables and connectors. Coaxial cables and connectors in accordance with MIL-STD-242 shall be provided for the RF signal lines of the notch filter assemblies to enable convenient installation within the notch filter units. Bayonet type connectors are preferred. Interface connectors for the notch filter units shall be:

RF signal	Panel jack M39012/18-0001 Cable plug M39012/16-0002
DC power	Mounting receptacle MS3112E-14-4P Cable plug MS3116F-14-4S

Four additional cables, each three feet (91.44cm) ±.25 inch (6.35mm) long, shall be provided for each notch filter unit signal leads.

3.2.15 Electrical power wiring. Electrical power wiring shall be provided to connect the ±15 VDC of the individual filter assembly switches to a common DC power interface connector for the notch filter unit.

3.3 Physical characteristics. A notch filter unit shall contain a set of eight notch filter assemblies and an associated switching network, and shall conform to the enclosure requirements of Sanders Associates Installation Drawing 1047702 for mounting into a standard nineteen-inch cabinet rack. In addition, the unit chassis shall have provisions for mounting government furnished heavy duty slides, shall weigh less than 160 pounds, and the cables shall exit at the chassis rear. Adequate space shall be provided between the notch filter unit and the cabinet to prevent any need for sharp bends in the cables.

3.4 Reliability. The specified mean-time-between-failure (MTBF) of a notch filter unit, θ_o, as defined by MIL-STD-781, shall be at least 10,000 hours.

3.5 Maintainability. A notch filter assembly shall be designed to have a predicted mean corrective maintenance time (M_{ct}) of 1.0 hour or less. The maintainability prediction shall be in accordance with MIL-HDBK-472. The maximum corrective maintenance time ($M_{max\ ct}$), within which ninety percent of all repairs are to be accomplished, shall be 3.0 hours or less.

3.6 Temperature and humidity. The filter shall operate as specified under the environmental conditions specified in MIL-E-16400 for Range 4 equipment. Although phase tracking measurements shall be made over Range 4, the required performance is defined only for the 25°C ±5°C range. The notch attenuation requirement shall be maintained over temperature for a reference center frequency established at 25°C ±5°C.

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3.7 Design and construction. The notch filter assembly and its component subassemblies shall conform to all applicable requirements for MIL-E-16400 for materials, processes, design, construction, and workmanship unless otherwise specified herein.

3.7.1 Interchangeability. Notch filter assemblies shall be identical in size and external electrical and mechanical configuration to permit interchanging the assemblies in any of the assembly locations and within any notch filter unit concurrently conforming to the performance requirements of the specification.

3.7.2 Marking. The notch filter unit and assemblies shall be legibly and permanently marked with the manufacturer's federal supplier identification code number, manufacturer's part number, serial number, reference designation and identification for all input and output connectors.

3.7.3 Enclosures. The notch filter assembly shall conform to the enclosure requirements of MIL-STD-108 for dripproof up to 15 degrees inclination.

3.7.4 Safety. In order to achieve safe installation, operation, and maintenance, the equipment shall conform to the safety standards of MIL-STD-454.

3.7.5 Electromagnetic compatibility. The generation of electromagnetic interference by the equipment and the vulnerability of the equipment to electromagnetic interference, shall be controlled within the limits of MIL-STD-461, Class 1A, for the tests specified herein:

RE 02	Upper frequency limit:	400 MHz
RS 03	Upper frequency limit:	50 MHz

3.7.6 Synchros. Synchros shall not be used in the notch filters.

3.7.7 Dials and pointers. Dials and pointers shall have white faces with black numerals, graduations and lettering. The points shall be black.

3.7.8 Noise. Noise acceptance limits are not applicable to this specification.

3.7.9 Test measurement. Test provisions for the equipment shall conform to MIL-STD-454, Requirement 32, except as specified herein.

3.7.10 General purpose electronic test equipment (GPETE). GPETE for use at all levels of maintenance shall be selected from the Standard and Substitute Standard GPETE listed in MIL-STD-1364 (Navy) "Standard General Purpose Electronic Test Equipment."

3.7.11 Parts. Parts for the notch filter unit shall be military standard in accordance with the requirements of MIL-STD-242. Non-standard variable components may be used, if necessary, to conform to the performance requirements of the specification.

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3.7.11.1 Procedure for selection of electrical parts and material during design of military items. Prescreening of nonstandard parts approval requests shall be in accordance with MIL-STD-1631.

3.8 Shock and vibration. The equipment shall be capable of withstanding the shock requirements of MIL-S-901, Grade A, Type B, Class II. The requirements of MIL-E-16400 for resilient mount and mounting information plates shall not apply. The equipment shall be capable of withstanding the vibration requirements of MIL-STD-167/1, Type I.

3.9 Operating life. The notch filter unit shall have a minimum total operating life of 10 years without major overhaul but including reasonable servicing and replacement of parts.

3.10 Operational stability. The notch filter unit shall conform to this specification for continuous and intermittent operation without the necessity for readjustment of any controls which are inaccessible to the operator in normal use. Repositioning the notch shall not be necessary when subjected to the vibration requirements of MIL-STD-167/1, Type I, for a maximum vibration frequency of 25 Hertz (Hz) in accordance with MIL-C-24056.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of:

(a) Surveillance of the operation to determine that practices, methods, and procedures of the approved quality assurance program are being properly implemented in accordance with MIL-Q-9858.
(b) Government product inspection to measure quality of product to be offered for acceptance.
(c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until performance of product to prescribed criteria has been demonstrated.

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4.2 Classification of inspection. The methods of inspection fall within the following classifications:

(a) First article inspection (see 4.3)
(b) Production inspection (see 4.4)
(c) Inspection of preparation for delivery (see 4.6)

4.3 First article inspection. First article inspection shall consist of preperformance and performance tests performed on the first notch filter unit produced. The inspection shall at least include the tests and inspections of Table I, in accordance with MIL-E-16400 except as specified herein. Preperformance tests shall determine equipment compliance with the requirements of 3.2.1 through 3.2.13 under ambient conditions. Performance tests shall determine equipment compliance with the requirements of 3.2.1 through 3.2.8 and 3.2.10 under environmental conditions (temperature, humidity, shock and vibration). Equipment performance tests shall be performed prior to, once during each newly acquired equilibrium state and at the conclusion of each environmental test.

Table I. First article inspection

Inspection or test	Requirement	Reference
Surface examinations	3.1.1	MIL-E-16400
Preperformance	3.1.1	MIL-E-16400
Performance	3.1.1	MIL-E-16400
Temperature	3.6	MIL-E-16400
Humidity	3.6	MIL-E-16400
Shock	3.8	MIL-E-16400
Vibration	3.8	MIL-E-16400
Temperature test sequence		MIL-E-16400
Electromagnetic compatibility	3.7.5	MIL-E-16400

4.4 Production inspection. Before acceptance, production equipment shall have met the requirements of the contractor Quality Program Plan and shall have passed the inspections and tests of Table II, in accordance wtih MIL-E-16400 except as specified herein. Performance tests shall determine the equipment compliance with the requirements of 3.2.1 through 3.2.13, except 3.2.9, under ambient conditions.

Table II. Production inspection

Inspection or test	Requirement	References
Surface examination	3.1.1	MIL-E-16400
Performance	3.1.1	MIL-E-16400

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4.5 Inspection of preparation for delivery. Inspections shall be performed to verify compliance to Section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be as specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Preservation, packing, packaging and marking shall conform to MIL-E-17555, level C, unless otherwise specified herein.

6. NOTES

6.1 Intended use. The notch filter unit is to be used on Navy combatant ships to provide onboard special purpose receivers with antenna input overload protection from local transmitters. The filters will pass signal in the 0.1 to 32.0 MHz range and provide selectable, tunable notches in the range of 2.0 to 24.0 MHz.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Marking, levels of preservation, packaging and packing required (see 5.1).

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC

(Project 5915-N097

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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